

Research on Substation Gantry Modeling Based on IFC Standard

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Abstract—BIM technology plays an important role in design and construction. IFC standard is a principal standard for BIM data exchange. In this paper, IFC standard is extended to substation gantry. The spatial and physical structure expression of substation gantry based on IFC is studied, and the data system of substation gantry information model is established. Substation gantry modeling based on IFC standard provides a method for data exchange of substation 3D model and will promote substation 3D design.

Keywords—IFC; substation; gantry; modeling; data exchange

I. INTRODUCTION

Building Information Modeling (BIM) is a complete information model that integrates engineering information, procurement information, construction information and operation and maintenance information at different stages into a single model [1]. The BIM concept and technology have swept the engineering construction industry in the world since its introduction. It has triggered unprecedented changes in the construction industry and has become an increasingly important focus of the global engineering construction industry. BIM technology enables engineers and technicians to correctly understand and respond to various building information, providing a basis for collaborative work between design teams and all parties involved in building operations. It improves production efficiency, saves costs and shortens construction time [2,3].

BIM is the main technology of information integration, and industry foundation classes (IFC) are specifications for information description and data exchange. The IFC standard was developed by the International Alliance for Interoperability and is widely accepted and applied in the field of construction engineering. Without the IFC standard, BIM cannot be a versatile and broad technology. IFC is a computer-represented building data representation and exchange standard. Its goal is to provide a neutral mechanism that is independent of any specific system and suitable for describing product data throughout the life of the building project. The IFC standard can effectively support data exchange between various application systems in the construction industry and data management throughout the life of the building [4].

At present, most of the applied research objects for IFC standards and BIM technologies are concentrated in the field of architectural engineering, and there is a lack of research on substation information models and modeling methods based on IFC standards. Based on the existing IFC data standards, this paper selects substation gantry as the research object, creates the spatial structure expression and physical structure expression of substation gantry based on the IFC standard, and establishes the data system of the substation gantry based on the IFC data format.

II. SUBSTATION GANTRY DATA SYSTEM BASED ON IFC STANDARD

A. Substation Gantry Data System Extension

This section will focus on the main components of the substation gantry, designing and creating new IFC description entities. This can avoid redundant or out-of-range classes of the constructed framework information model and improve the efficiency of data exchange. The basic components of the substation structure are beams, columns, steel plates, ground columns, ladders, foundations, etc. Based on the IFC standard, the extension of the framework data description is designed to include the physics, spatial information of substation gantry, corresponding attribute sets and their relationships, as shown in Figure 1.

A spatial structure element (`IfcSpatialStructureElement`) is the generalization of all spatial elements that might be used to define a spatial structure. Besides `IfcSite` for site, `IfcBuilding` for building, `IfcBuildingStorey` for storey, `IfcSpace` for space, `IfcCivil` is added to provide a civil structure space [5].

`IfcElementComponent` is a small object attached to or contained in a component, which plays an auxiliary role in strengthening or connecting components. Generally speaking, these small objects do not need much attention, but they do play an important role. All kinds of fasteners and fittings are typical `IfcElementComponent`. Here `IfcGantryElementComponent` is created to contains components in gantry.

An `IfcCivilElement` is a generalization of all elements within a civil engineering works. It is added in IFC4 version, which is subtype of `IfcElement`. `IfcGantryElement` is created as a subtype of `IfcCivilElement`. It contains components of the substation gantry, such as base (`IfcGantryBase`), beam

(IfcGantryBeam), column (IfcGantryColumn), plate (IfcGantryPlate), ladder (IfcGantryLadder), groundrod (IfcGantryGroundRod). The extended part uses the default IFC framework mechanism to maximize model compatibility and consistency.

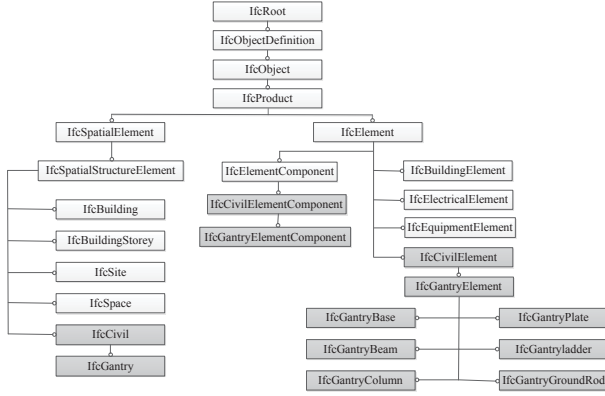


Figure 1. Substation gantry data system extension

B. Expression of space structure of substation gantry based on IFC standard

IfcSpatialStructureElement is a superclass of all spatial structures in IFC, breaking the project model into individual easy-to-operate sub-sets based on spatial layout. Although the IFC standard already has spatial description classes such as IfcBuilding, IfcBuildingStorey, IfcSite, and IfcSpace, they are all designed for building projects [6]. Therefore, the IfcCivil class is added to describe the spatial structure of civil structures. The class contains a description of the space and acts as a container for the civil artifacts. Its subclass IfcGantry describes the spatial structure of gantry. The IfcCivil can contain multiple IfcGantry entities, representing multiple monolithic structures of the gantries in a project. Its logical relationship is shown in Figure 2.

The IfcGantry class is used to provide characterization data for the substation gantry itself, including location, spatial geometry, height, and more. The basic characteristics of the type and height of the substation gantry system are expressed by attributes.

The hierarchical relationship between the architectural elements is linked by the relational entity IfcRelAggregates. Instances of entities such as IfcProject, IfcCivil are associated with the same hierarchy through IfcRelAggregates. The horizontal and vertical hierarchical relationship of the substation frame space structure is shown in Figure 2. IfcCivil is located in the global coordinate system of IfcProject, and the IfcGantry project is located in the civil space.

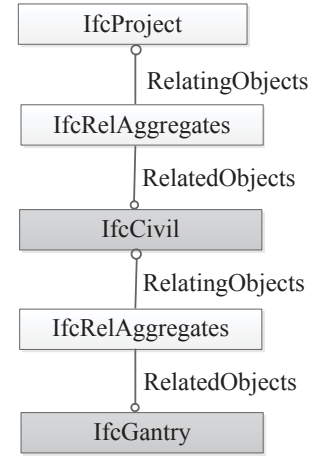


Figure 2. Hierarchical spatial relations of substation gantry

C. Expression of physical structure of substation gantry based on IFC standard

IfcElement is a superclass representing the physical structure elements in the IFC standard. The substation gantry physical elements are concentrated under the subclass IfcCivilElement of IfcElement. IfcGantryElement is an abstract superclass of all gantries, mainly containing the physical elements of the main components of the gantry.

In the IFC standard, in addition to having some static attributes, the specific requirements for different projects can be supplemented and improved by dynamic extensible attributes and their set IfcPropertySet. The attribute set is associated with the body entity through the relational entity IfcRelDefinedByProperties.

In the IFC standard, a relative coordinate system is used to position the components. The position of the component is referenced to the coordinate system in which the space is located. For example, positioning a beam, described by ObjectPlacement, defines the local coordinate system and the reference coordinate system. The attribute PlacementRelTo refers to the reference coordinate system, and this reference coordinate system is the coordinate system in which the gantry is located. IfcGantry is based on the coordinate system of IfcCivil, and IfcCivil defines the overall coordinate system, and there is no reference system. RelativePlacement is a coordinate transformation in a relative coordinate system, that is, a local coordinate system is defined. This is the case for multiple layers of reference. Under the premise of clarifying the hierarchical relationship of each component, the stepwise query is performed to determine the absolute coordinates of the component [7].

The relational entity IfcRelContainedInSpatialStructure is used to establish a relationship between a physical structure element and its own spatial structure element. A physical element can only be included in one spatial element. In addition, the position attribute IfcLocalPlacement of the physical element refers to the position attribute of the spatial element containing the physical element, and the position attribute of the spatial element always refers to the position attribute of the superior space element until the spatial structure

element having the global absolute coordinate. Through this layer coordinate conversion, the position attribute expression of the physical element is made more concise and fine.

The IFC standard was developed and gradually refined with reference to the STEP standard and is defined in the EXPRESS language. In an IFC file, any entity (such as IfcGantryElement) describes its own information through attributes. Attributes are divided into direct attributes, inverse attributes, and exported attributes. The direct attribute is the index quantity or direct information, such as GlobalID, Name; the derived attribute refers to the attribute expressed by other entities, such as OwnerHistory, ObjectPlacement; the inverse attribute refers to the attribute linked by the associated entity, such as HasAssociations through the associated entity IfcRelAssociates can associate material information for a component. The attributes of the IFC entity are obtained through the inheritance relationship. For example, the component IfcGantry has 32 attributes in the IFC4x2 version, and only has the PredefinedType attribute. The remaining 31 attributes inherit the attributes from the IfcRoot to the IfcCivilElement. IfcGantry is the subtype of the IfcCivilElement, and the previous IfcElement is the supertype of the IfcCivilElement. By analogy, the top layer is IfcRoot. In the IFC physical file, the statement IfcGantry only displays 9 attributes, including direct attributes and exported attributes. The remaining 23 attributes are inverse attributes, and the corresponding relationship is shown in Figure 3.

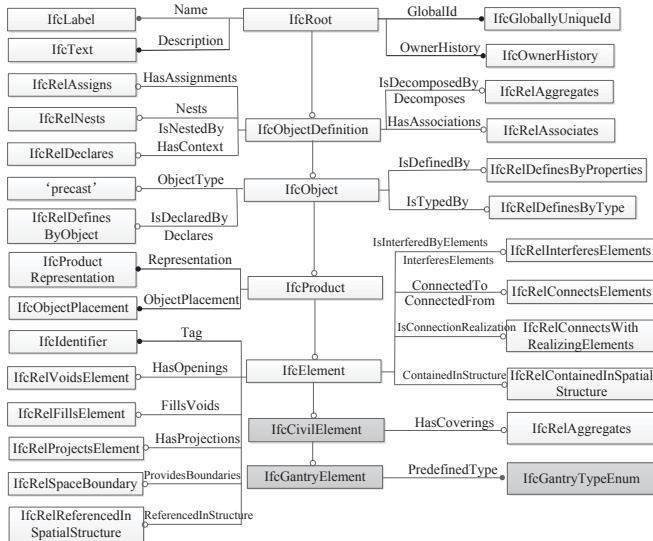


Figure 3. Attribute of substation gantry in IFC standard

III. SUBSTATION GANTRY MODELING

According to the definition of each spatial and physical structural element in the extended framework IFC standard, a 3D model of the framework is constructed, and the corresponding IFC entity is created by using the geometric shape and position attribute of the component to form a framework information capable of accurately describing and displaying the component information model.

There are many types of substation gantries, and different modeling methods are required for different types of gantries. In order to reduce the modeling workload and shorten the modeling cycle, parametric modeling technology can be used to build a template library for different components of the substation gantry system to form a common gantry model library.

1) Basic data collation. Design materials and other documents of the framework should be organized and read in order to obtain information that helps to determine the expression of the 3D model of the component.

2) Analyze component attribute parameters. There are generally two types of parameters describing component attributes: deterministic, more uniform, and more frequently used parameters; no deterministic, too many changes and irregularities, parameters that cannot be uniformly expressed using parameters, need to be used in the model. The time is determined by the modeler based on actual conditions. For complex components, it can be divided into several sub-components, respectively, sub-models are built and organically assembled to build a nested model.

3) Generate a parametric model. After the parameter design is completed, the framework of the model is basically determined, and a parametric model is generated.

The parameter modification method must be supported by the model library. The model library is usually manually created by the user and saved in the directory pointed to by the program. When you need to use it, you can open the model file from the model library, modify and rebuild the specified size parameters, and you can get the model that meets your needs. The program design of this method is small, and has nothing to do with the modeling process. It is suitable for situations where the model is highly standardized or the modeling process is complicated and the variable parameters are small. The parameter modification method modeling has higher requirements on the model library. When manual modeling, it is necessary to comprehensively consider the dimensioning method. The association and constraint relationship between the size parameters should be avoided. The size parameters that need to be modified must be independently labeled. The name of the dimension can be passed. Check out its properties. The input of variable parameters can be realized by various methods such as dialog box and database, or can be obtained from the design calculation result of the program. Before the model parameters are modified, the corresponding parts in the parts library must be opened. The operation of opening the parts can be done manually. It can also be done through the program. This modeling method does not require the programmer to master a large number of API modeling functions, and the complexity of the program is independent of the complexity of the model. Since the parameter modification must correspond to the specified label object and part object, this modeling method has strong dependence on the model library. Once the model library is determined, it cannot be modified at will, otherwise the program will not work properly. The modeling process of substation gantry is shown in Figure 4.

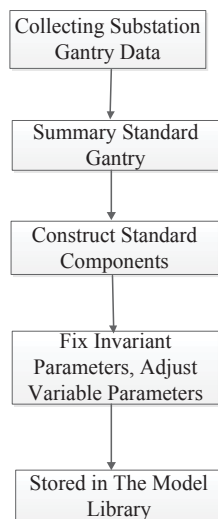


Figure 4. Modeling process of substation gantry

IV. CONCLUSIONS

The data structure of the substation gantry is extended based on the IFC standard. The spatial structure and physical structure of the gantry information model are studied. The model attributes of the gantry are clarified, and the parametric modeling method of the substation gantry is applied. The standard model library of gantry is proposed. The establishment of the substation gantry information model will

contribute to the exchange of substation 3D design data and the cross-platform flow of information.

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